

Module 2 Lab 1: Python Warm-Up + First Push

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This notebook works through Steps 1–6 of the Module 2 Lab 1 assignment: core Python constructs, a dictionary, and loading/inspecting `chickwts.csv` with pandas.

Step 1: Wages with f-strings

```
In [1]: name = "Azra"
        hourly_wage = 32.00
        hours = 20
        weekly_pay = hourly_wage * hours
        print(f"{name} earns {weekly_pay} per week")
```

Azra earns 640.0 per week

Step 2: Count the warm days

```
In [2]: temps = [61, 65, 72, 58, 70, 74, 68]
        warm_days = 0
        for t in temps:
            if t > 65:
                warm_days = warm_days + 1
        print(warm_days)
```

4

Step 3: Write a temperature converter

```
In [3]: def to_celsius(f):
        return (f - 32) * 5 / 9

        print(round(to_celsius(temps[0]), 1))
```

16.1

Checkpoint: Steps 1–3 ran top to bottom with no `NameError`, and the warm-day count matches the expected value of 4.

Step 4: A movie dictionary

```
In [4]: movie = {"title": "Inception", "year": 2010, "rating": 8.8}

for key, value in movie.items():
    print(key, ":", value)
```

```
title : Inception
year : 2010
rating : 8.8
```

Step 5: Load and inspect chickwts.csv

```
In [5]: import pandas as pd

df = pd.read_csv("chickwts.csv")
df.head()
```

```
Out[5]:
```

	weight	feed
0	179	horsebean
1	160	horsebean
2	136	horsebean
3	227	horsebean
4	217	horsebean

```
In [6]: df.shape
```

```
Out[6]: (71, 2)
```

```
In [7]: df.describe()
```

```
Out[7]:
```

	weight
count	71.000000
mean	261.309859
std	78.073700
min	108.000000
25%	204.500000
50%	258.000000
75%	323.500000
max	423.000000

Step 6: Interpret describe()

The `describe()` output shows 71 chicks with a mean weight of about 261 grams and a median (50th percentile) close to that, so the distribution isn't badly skewed. The spread is wide, though: weights range from about 108 grams to 423 grams, with a standard deviation near 78 grams, meaning individual chicks vary a lot around the average. That's not surprising once you remember `weight` is pooled across six different feed types — the variation likely reflects real differences in how effective each feed was, not measurement noise, which is exactly what you'd want to check next by grouping `weight` by `feed`.

Appendix: AI Use Disclosure

Tool used: Claude (Anthropic), via Claude Code / Cowork.

How it was used: Drafted the code cells for Steps 1–5 following the patterns given in the lab's tutorials, and produced a first-draft interpretation for Step 6 that I reviewed and confirmed reflects the actual `describe()` output for this dataset.

Key prompts:

- "Build the Module 2 Lab 1 notebook (steps 1–6) following the tutorial patterns exactly, using `chickwts.csv`."
- "Write a Step 6 interpretation of `describe()` for the chick weights."